Project Proposal

Project Title - Collabora: A Smart Project Collaboration and Management Platform for University Students

1. Introduction

Group projects are an essential part of university education because they help students develop teamwork, communication, leadership, and software development skills. However, managing a group project efficiently is often difficult. Students usually rely on multiple applications such as Messenger for communication, Google Drive for file sharing, Trello for task management, and Google Calendar for scheduling. Using several different platforms often causes confusion, poor communication, missed deadlines, duplicate work, and difficulty tracking each member's contributions.

The proposed system, Collabora, aims to provide an integrated web-based platform where students, team leaders, and lecturers can collaborate efficiently throughout the entire project lifecycle. The platform combines project management, task assignment, file sharing, communication, progress tracking, meeting management, and reporting into a single system.

This project will be developed using Spring Boot (J2EE) to implement RESTful web services, authentication, database management, and business logic. Spring Boot provides a secure, scalable, and maintainable architecture suitable for developing enterprise-level web applications.

2. Problem Statement

Many university project teams experience various challenges while completing group assignments. Some common problems include:

- Project information is scattered across multiple applications.
- Team members are unclear about their assigned responsibilities.

- Deadlines are easily forgotten or missed.
- Project progress is difficult to monitor.
- Files are often duplicated or misplaced.
- Communication between members is inconsistent.
- Lecturers have limited visibility into the progress of each project.
- Team leaders cannot accurately evaluate each member's contribution.
- There is no centralized platform specifically designed for managing academic group projects.

These problems reduce productivity, create misunderstandings among team members, and make project management more difficult.

Therefore, there is a need for an integrated collaboration platform that enables students and lecturers to manage projects more effectively from project creation to final submission.

3. Project Objectives

General Objective

To develop a web-based Project Collaboration and Management Platform that improves communication, task management, collaboration, and project monitoring among university students and lecturers.

Specific Objective

Develop a secure user authentication and authorization system using Spring Security and JWT.

- Allow lecturers to create and manage academic projects.
- Enable students to collaborate within project teams.
- Provide task assignment and task tracking functionalities.
- Implement milestone and deadline management.

- Support file uploading and centralized document management.
- Provide real-time notifications for important project activities.
- Track project progress automatically using task completion.
- Record member activities for contribution analysis.
- Generate reports that help lecturers monitor project performance.

4. Scope

The proposed system focuses on managing academic group projects from project creation until project completion.

Functional Scope

The system will provide the following functions:

User Management

- User registration and login
- Role-based access control
- User profile management

Project Management

- Create projects
- Edit project information
- Delete projects
- Assign supervisors
- Create project teams

Team Management

- Invite members
- Manage team members

- Assign team leader

Task Management

- Create tasks
- Assign tasks
- Set priorities
- Set deadlines
- Update task status
- Track task completion

Milestone Management

- Create milestones
- Assign deadlines
- Monitor milestone completion

File Management

- Upload project files
- Download files
- Organize documents
- Maintain file versions

Communication

- Task comments
- Project discussions
- Activity timeline
- Notifications

Meeting Management

- Schedule meetings

- Store meeting minutes
- Share meeting links

Dashboard

Students can view:

- Assigned tasks
- Upcoming deadlines
- Team progress
- Notifications

Lecturers can view:

- Project status
- Team progress
- Member contributions
- Overall project analytics

Reports

- Weekly project reports
- Task completion reports
- Member contribution reports
- Progress summaries

System Limitations

The first version of the system will not include:

- Video conferencing
- Online code editor
- Automatic code evaluation
- AI-based task recommendations

- Mobile application
- Third-party project management integrations

These features may be considered for future enhancements.

5. Technologies

The proposed system will be developed using the following technologies.

Category	Technology
Programming Language	Java 21
Backend Framework	Spring Boot
Security	Spring Security, JWT
Database	MySQL
ORM	Spring Data JPA (Hibernate)
Build Tool	Maven
Frontend	React.js
UI Framework	Tailwind CSS
API Testing	Postman
Version Control	Git & GitHub

IDE	IntelliJ IDEA / VS Code
Database Tool	MySQL Workbench
File Storage	Local Storage (Cloud Storage in future)
Documentation	Swagger/OpenAPI (Optional)

6. Expected Outcomes

Upon successful completion of the project, the proposed system is expected to provide the following outcomes:

- A centralized platform for managing university group projects.
- Improved communication among project members.
- Better task assignment and progress monitoring.
- Reduced risk of missing deadlines.
- Efficient project document management.
- Increased transparency of individual member contributions.
- Better supervision for lecturers through real-time project monitoring.
- Improved collaboration and productivity among students.
- Practical implementation of Spring Boot technologies in an enterprise-style application.

Additionally, the project will demonstrate the application of modern software engineering principles such as layered architecture, RESTful APIs, authentication, authorization, database management, and collaborative system design.

7. Conclusion

The Collabora: Smart Project Collaboration and Management Platform is proposed to address the common challenges faced by university students during group project development. By integrating project management, task assignment, communication, document management, scheduling, and progress tracking into a single platform, the system will improve collaboration, increase productivity, and simplify project supervision.

The project also provides an opportunity to apply enterprise-level Java development technologies using Spring Boot, Spring Security, JWT authentication, MySQL, and React. The successful completion of this project will not only benefit students and lecturers but also demonstrate practical knowledge of full-stack software development and modern web application architecture.